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Influence of lithium plating on lithium-ion battery aging at high temperature

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ABSTRACT

Lithium plating is an important issue for lithium-ion battery safety and cycle life that can be caused by cycle at low temperature. In this study, we investigated battery aging over an extended low-temperature cycle and at high temperature after the low-temperature cycle. Nondestructive and destructive analysis methods were used to study battery electrochemical dynamics and changes to the materials, and the results indicated that battery aging over the extended low-temperature cycle could be divided into two stages. The mechanism of the first stage is that of "dead lithium," which causes significant capacity fading, whereas that of the second stage is solid electrolyte interphase (SEI) growth. It was also found that plated lithium accelerates battery aging at high temperature; part of it is intercalated into anode, another part of it reacts with the electrolyte, thus consuming the plated lithium. Thus, electrolyte decomposition and SEI growth are the primary aging mechanisms at high temperature after low-temperature cycling. The plated lithium content and charge/discharge rate affected plated lithium consumption during the high-temperature cycle. Plated lithium was not depleted for the battery with significant amounts of plated lithium or upon charging with high current and discharging with both high and low current when the state of health (SOH) decreased to 70%. The factors affecting battery aging ranked in order of importance are charge current > plated lithium content > high temperature > discharge current.

1. Introduction

Lithium-ion batteries are powerful and energetic energy storage devices that are widely used in mobile electronics, grid systems, and electrical vehicles [1–3]. Thus, cycle life and safety issues are important [4–9]. The cycle life of a battery determines its economic value to some extent, which influences the promotion of lithium-ion batteries. It is therefore important to study the aging mechanisms of Li-ion batteries and the factors that influence battery aging as the results can be used to better predict and prolong battery cycle life. Lithium plating affects both cycle life and the safety of lithium-ion batteries as it consumes reversible lithium and may induce internal short-circuits. The influence of plated lithium on battery safety has been studied [1,10,11], and various methods for suppressing lithium plating and lithium dendrite growth have been developed to ensure battery safety, such as Janus separators, graphene oxide, etc. [12–15]. However, the evolution of aging mechanisms during extended low-temperature cycling and the influence of plated lithium on battery aging at high temperature, which are important to battery cycle life, have not been studied in detail. To fill this gap, we investigated battery aging at high temperature after extended low-temperature cycling.

It is reported that the main battery aging mechanisms include loss of lithium ions (LLI) and loss of active material (LAM) [16,17]. LLI is mainly caused by electrolyte decomposition, solid electrolyte interphase (SEI) growth, and lithium plating. LAM is mainly caused by graphite exfoliation, transition metal dissolution, destruction of the electrode structure, and current collector corrosion [17,18]. In most cases, LLI induces battery capacity fading for NMC batteries [16]. Goodenough [19] reported that the "stable window" of an electrolyte is the energy difference between the lowest unoccupied molecular orbital and the highest occupied molecular orbital. If the electrochemical potential of the anode is above the lowest unoccupied molecular orbital, the electrolyte will be reduced, whereas if the electrochemical potential of the cathode is below the highest occupied molecular orbital, the electrolyte will be oxidized [19]. It is reported that if the intercalation potential of lithium into graphite is below the reduction potential of the organic electrolyte, the electrolyte decomposes to form SEI [20]. The electrolyte also decomposes on the surface of the cathode to form a cathode

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electrolyte interphase (CEI). SEI and CEI have high lithium-ion conductivities and negligible electronic conductivities, and thus their electronically insulating properties inhibit further electrolyte decomposition [20]. So, these decomposition mechanisms are often kinetically delayed by the presence of SEI and CEI layers. Charging with high voltage accelerates SEI growth and capacity fading of lithium-ion batteries. Based on equilibrium thermodynamics, the occurrence of lithium plating will typically occur at potentials below 0C vs Li/Li+, due to the overpotentials associated with charging [3,21–23]. Increased impedance causes a high overpotential. Therefore, based on equilibrium thermodynamics, an increase in anode impedance accelerates lithium plating [24]. "Dead lithium" and SEI growth occur after lithium plating, both of which increase the impedance of lithium-ion batteries. It is reported that lithium plating is self-accelerated during battery cycling at low temperature because the resistance of the anode increases after the thickening of the SEI film and consumption of the electrolyte caused by the reaction between the plated lithium and the electrolyte [23]. However, the experimental results of Tao [21] indicated that the onset of lithium plating occurred only when the surfaces of the highly oriented pyrolytic graphite particles were fully lithiated. The graphite particles were overcharged until lithium plating occurred. Voltages of graphite below 0 V vs. Li/Li^+ may not result in the onset of lithium plating [21]. Therefore, lithium plating will not necessarily occur if the anode potential is modestly lower than 0 V. However, lithium plating may occur continuously when the battery is cycled at low temperatures. If it is self-accelerated, the question is whether the battery aging rate will increase with increased cycling at low temperature. This deserves further study.

The operation conditions of lithium-ion batteries are complex, and lithium-ion battery aging is influenced by many stress factors, such as ambient temperature, charging/discharging rate, and charging and discharging cut-off voltage [25]. Su et al. [25] conducted orthogonal experiments to study the main factors influencing battery aging. A $LiNi_xCo_yAl_xO_2$ battery (2.85 Ah) with graphite as anode was used in their study. Their results indicated that the factors that influenced battery aging in order of importance were charge current > charge cut-off voltage > ambient temperature > discharging durations > discharging cut-off voltage > discharge current > charging durations. However, the shortcoming of orthogonal design is that the interactions between these factors are ignored. It is also reported that high charging voltage and high charging/discharging rate accelerate side reactions and battery aging [26]. The thermal and nonthermal effects of discharging rate on lithium-ion battery aging were decoupled in a study by Xu et al. [27]. A $LiCoO_2$-$Li(NiCoAl)O_2$ battery with graphite as anode was used in their study. The surface temperature of the battery was controlled at 45 ± 1 °C. Their results indicated that the thermal and nonthermal effects of discharge rate accelerated SEI growth and particle cracking, respectively. Thomas et al. [28] studied the influence of ambient temperature on lithium-ion battery aging. Commercial 18,650-type $Li_xNi_{1/3}Mn_{1/3}Co_xO_2/Li_yMn_2O_4$ batteries with graphite/carbon as anode were cycled with 1C at different constant temperatures. Their results indicated that SEI growth was the primary aging mechanism at high temperature (>25 °C). However, lithium plating was the primary aging mechanism during low-temperature cycling (<25 °C), and both high and low temperatures accelerated battery aging compared to cycling at 25 °C. It was even assumed that SEI growth could be ignored for a cylindrical 26,650-type $LiFePO_4$ commercial battery with graphite as anode cycled at temperatures ranging from −26 to 20 °C [29].

Lithium plating thus accelerates battery aging and causes safety issues. Most studies have focused on the onset of lithium plating and its detection [3,30]. However, lithium plating may occur continuously throughout the low-temperature cycle. Petzl et al. [31] studied battery cycling aging at −22 °C with 1C. A cylindrical lithium–iron–phosphate (LFP) 26,650-type cell with graphite as anode was used in their study. Their results indicated that an inflection point occurred on the capacity curves, after which the capacity fading rate decreased significantly for NMC battery. The initial occurrence of lithium plating inhibits further lithium plating. Zhang et al. [32] reported that the capacity curve of a battery cycled at −10 °C showed an inflection point. A large-format prismatic commercial battery with a layered metal oxide of NMC523 and a mixture of graphite and carbon black as the cathode and anode was used in their study. The battery was cycled at −10 °C with 2C, and the amount of lithium plating increased rapidly before the inflection point. After that, it increased more slowly. However, the reason for the occurrence of an inflection point deserves further study. Existing studies on the aging mechanisms of lithium-ion batteries have been conducted under constant operating conditions, i.e., constant temperature, constant charging/discharging rate, and constant charging and discharging cut-off voltage. However, temperature has a significant influence on battery aging. The ambient temperature in the summer and winter or in the southern and northern parts of China varies, and thus the aging mechanism at high and low temperature is different. What are the aging behavior and mechanisms for lithium-ion batteries cycled at high temperature after being cycled at low temperature [i.e., the battery is cycled to a 90% state of health (SOH) at low temperature and then at high temperature to the end of its life]? To the best of our knowledge, this has not been studied until now.

The SOH is defined as follows:

$$SOH = \frac{Q_c}{Q_i} \tag{1}$$

where Q_c is the capacity obtained by capacity tests after cycling, and Q_i is the initial capacity of the battery.

In this study, we investigated the aging behavior and mechanisms of a lithium-ion battery cycled at high temperature after being cycled at low temperature (i.e., the battery was cycled to 90% SOH at low temperature and then at high temperature to the end of its life). First, the aging behavior of a battery subjected to an extended constant low-temperature cycle and the aging mechanisms were studied, and the reason for the decreased capacity fading rate is discussed. Second, after being cycled to 90% SOH at low temperature, the aged battery was cycled at high temperature. The aging behavior, mechanisms, and the influence of lithium plating on battery aging at high temperature were studied. Finally, the influence of plated lithium content and charging/discharging rate of the high-temperature cycle on aging at high temperature were studied. The experimental results of this manuscript are useful for the design of battery management systems that will prolong the cycle life and safety of batteries.

2. Experimental

Commercially available 2.5 Ah 18,650-type cylindrical batteries (from Far East Battery, Yichun, Jiangxi province, China) were used in this study. Their charging and discharging cut-off voltages were 4.2 V and 2.75 V, respectively, and their standard charging and discharging rates were 0.5C and 1C, respectively. The cathodes comprised 95% Li $(Ni_{0.5}Co_{0.2}Mn_{0.3})O_2$ and 5% $LiMn_2O_4$. The anode was graphite. The recommended operating conditions were as follows: charge, 0–45 °C; discharge, −20–60 °C; maximum charge current, 250 mA at 0 °C; maximum discharge current 2500 mA at −20 °C. The batteries were cycled three times before testing. The cycle modes were constant current–constant voltage (CC–CV) charging and CC discharging. During CC charging mode, the batteries were charged to 4.2 V with 0.5C. The voltage was constant after that. The charging was stopped until the battery current decreased to 25 mA (0.01C). The batteries were discharged to 2.75 V with 0.5C during CC discharging mode. After that, reference performance tests (RPTs), including capacity and electrochemical impedance spectroscopy (EIS) tests were conducted to obtain the initial capacity and impedance of the batteries. Capacity testing included CC–CV charging mode with 0.5C and CC discharging mode with 0.5C. The EIS of the batteries was obtained using an

electrochemical workstation. A potentiostatic mode at an amplitude of 5 mV in the frequency range of 0.01–10^6 Hz was used. Before the tests, the batteries were charged to 100% SOC with CC–CV mode.

2.1. Aging tests

After the RPTs, the lithium-ion batteries were put into a climate test chamber at −10 °C and cycled with 0.4C CC–CV charging–0.4C CC discharging mode. There was a 30-min rest period between CC–CV charging and CC discharging. The RPTs were conducted after several cycles, and the cycles were terminated when the SOH of the batteries decreased to a predetermined value. Afterwards, the aged lithium-ion batteries were placed into a thermal chamber at 50 °C and cycled in CC–CV charging–CC discharging mode with different charging/discharging rates. The experimental design is presented in Table 1. The RPTs were conducted when the SOH of the batteries decreased to predetermined values (90%, 80%, and 70%). The aging tests were terminated when the batteries reached their end of life (70% SOH). More than two batteries were used under most experimental conditions. The differential voltage (DV) and incremental capacity (IC) derived from cycle and capacity tests and EIS were used to nondestructively analyze the aging mechanisms of the lithium-ion batteries.

2.2. Postmortem analysis

To obtain more information about the aging mechanisms of the batteries, lithium-ion batteries with different SOH values were disassembled in a glovebox to conduct postmortem analysis. Before disassembly, the lithium-ion batteries were discharged to 2.75 V with 0.5C. After disassembly, the electrode materials were harvested from the "jelly roll" and washed in dimethyl carbonate (DMC) for 1 min as suggested by Ref. [33]. After washing, they were placed in a vacuum environment for hours to evaporate the DMC. X-ray diffraction (XRD; Rigaku Smartlab), scanning electron microscopy (SEM; ZEISS Gemini 300), X-ray photoelectron spectroscopy (XPS; Thermo Scientific K-Alpha), and energy-dispersive spectroscopy (EDS; ZEISS Gemini 300) were then used to analyze the structure, surface topography, surface composition, and element changes, respectively, of the electrode materials.

3. Results and discussion

3.1. Battery aging at low temperature

The aging behavior of the lithium-ion battery during extended low-temperature cycling is depicted in Fig. 1(a). It can be seen from the figure that battery aging can be divided into two stages. In the first stage, SOH decreases quickly and linearly with increasing cycling. In the second stage, the aging rate decreases. SOH also decreases linearly with increasing cycling. Incremental capacity analysis (ICA) and DV were used to nondestructively analyze lithium plating during cycling at low temperature as they are powerful tools for detecting lithium plating [29]. The results are derived from the charging and discharging data of batteries and can reflect phase changes and redox reactions during the charging–discharging process [17,20,21]. After lithium plating, some of the plated lithium can be stripped during discharging. This is considered reversible lithium plating. However, another part of the plated lithium loses contact with the graphite and becomes "dead lithium." This is considered irreversible lithium plating. The electrolyte can also react with the anode to form SEI. Capacity fading is caused by "dead lithium" and SEI growth. Owing to lithium stripping, an additional phase transition occurs during discharging, as indicated by the voltage plateau on the voltage curves and the additional peak on the IC and DV curves during discharging. The IC and DV curves derived from discharging at low temperature are shown in Fig. 1(b) and (c), respectively. The discharging curves are shown in Fig. S1. Peak "n" in Fig. 1(b) and (c) reflects lithium stripping, and it disappears after the 16th cycle. Fig 1(c) shows that the reversible capacity decreases from 0.43 Ah to 0.02 Ah from the 1st to the 11th cycle. This means that lithium stripping decreased continuously to zero with extended cycling at low temperature. The IC curves derived from the charging of the capacity test are depicted in Fig. 1(d). Peaks (1)–(3) represent last lithium interaction reaction into anode, the NCM phase change, and the NCM solid solution, respectively [16]. It can be seen from Fig. 1(d) that peak (1) initially increases and then decreases with increased cycling and that peaks (1)–(3) shift to higher voltages. This means that LLI and LAM at the positive electrode are the main aging mechanisms [16]. Peak (1) decreases significantly after the 29th cycle. Because peak (1) is closely correlated with LAM at the anode [16], LLI and LAM at the anode are the main aging mechanisms after the 29th cycle.

EIS reflects the electrochemical dynamics of lithium-ion batteries during charging and discharging. It is also a powerful tool for analyzing aging mechanisms. The results of the EIS tests for lithium-ion batteries cycled at low temperature are shown in Fig. 1(e). The EIS plots can be divided into high-, mid-, and low-frequency regions. The test results in the high-frequency region reflect conduction through the wires, electrolyte, and separator. The test results in the mid-frequency range are shown as two semicircles. The first semicircle in the high mid-frequency region reflects SEI impedance. The second semicircle in the low mid-frequency region represents charge-transfer resistance. The test results in the low-frequency regions reflect diffusion limitation in the solid phase. The intersections of the plots with the axis represent the ohmic resistances of the wire, current collector and electrolyte [1,34,35]. It can be seen from Fig. 1(e) that the two semicircles show less change before the 15th cycle. After that, the two semicircles expand, especially the low mid-frequency semicircle. This means an increase of SEI and charge-transfer resistance, which reflects serious side reactions on the anode [36].

Fig. 1(a) shows that aging rate of the lithium-ion battery decreases after the 29th cycle. This means that total capacity fading caused by "dead lithium" and SEI formation decreases. Fig. S2 indicates that the CV time during charging at low temperature increases quickly up to the 20th cycle. Because charging with high voltage accelerates electrolyte decomposition, SEI formation increases as the CV time increases. It can be concluded that the amount of "dead lithium" decreases with increasing cycling. Fig. 1(b) and (c) show that the amount of reversible lithium decreases. But it is reported that plated lithium can be intercalated into the graphite during CV and rest mode [21,36]. It can be seen from Fig. 1(b) and (c) and Fig. S2 that lithium stripping during discharging disappears after the 16th cycle and that the time for CV mode increases quickly up to the 20th cycle. It is expected that the increasing time of CV mode causes more intercalated lithium plating and SEI growth, whereas causes less lithium stripping during discharging. "Dead lithium" occurs only after stripping during discharging [29], which is why the amount of "dead lithium" decreases. Fig. 1(d) shows LAM at the anode after the 29th cycle. Fig. 1(e) shows that SEI and charge-transfer increases after the 24th cycle. This means that serious side reactions occurred on the anode. The primary aging mechanism of the battery during extended cycling at low temperature thus changed from lithium plating to SEI growth.

Table 1
Experimental design.

Batteries	SOH after low-temperature cycle (%)	Charging rate at high temperature (C)	Discharging rate at high temperature (C)
1	90	0.5	1
2	90	1	1
3	90	1.5	1
4	90	1	1.5
5	90	1	0.5
6	98	1	1
7	82	1	1

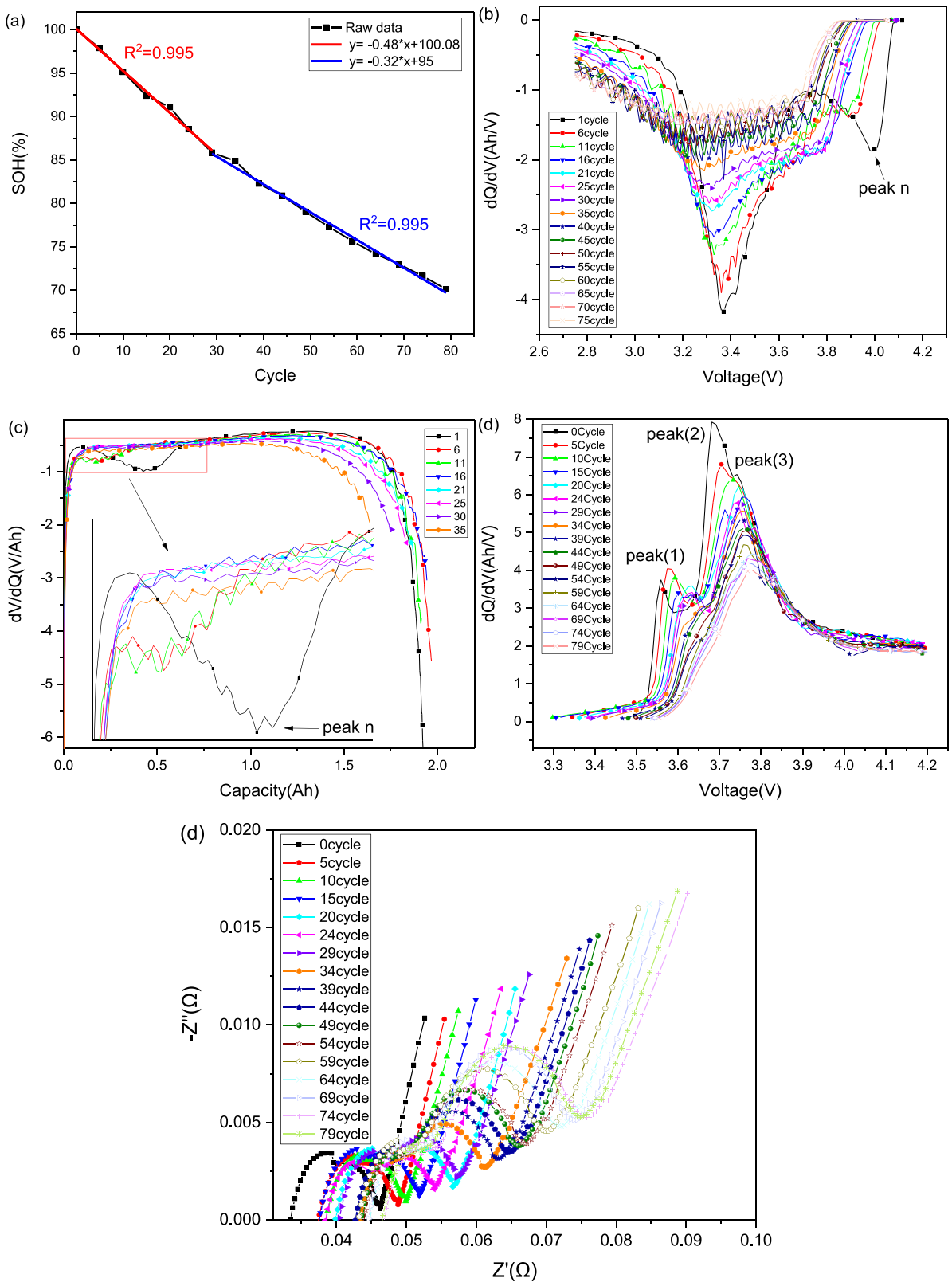


Fig. 1. Battery aging at low temperature. (a) SOH changes with cycle; (b) IC curves from cycle at low temperature; (c) DV curves from cycle at low temperature; (d) IC curves from capacity test; (e) EIS plots.

To obtain more information about the aging mechanisms of lithium-ion batteries, SEM, EDS, XRD, and XPS analyses were conducted to analyze the changes in the electrode materials surface, surface element, structure, and surface composition, respectively. Optical images of the electrode materials from the disassembled batteries are shown in Fig. S3. It can be seen from the figure that the cathodes of batteries with different SOH values under low-temperature cycling changed less compared to that of the fresh battery. However, plated lithium occurred on the

surfaces of anodes for batteries with different SOH values under low-temperature cycling. The anode is divided into a black area (part 1) and a white area (part 2), and the size of the white area increases with decreasing SOH.

The results of the SEM analysis are shown in Fig. 2. The fresh anode shown in Fig. 2(a) and (b) presents a sheet structure, whereas depositions cover the surface of the anode of the battery with 98.7% SOH, as shown in Fig. 2(c) and (d). Although the anode shown in Fig. 2(d) still presents a sheet structure, many bumps occur on the surfaces of the anodes with 90% and 70% SOH (part 1), as shown in Fig. 2(e–h). The anode of the battery with 70% SOH (part 2), shown in Fig. 2(i), shows a flat anode surface. Fig. 2(j) shows that needle-like materials uniformly cover the surface of the anode, indicating that the anode no longer presents a sheet structure. Fig. S4(b), (d), and (e) show that the bumps are the same as those of the anode of the battery with 70% SOH (part 2). The bumps and part 2 of the anode are plated lithium and the products of reactions between the electrolyte and the plated lithium. A more extensive white area indicates more plated lithium and side reactions. Fig. 2(k–n) show that some deposits cover the surface of the cathode of the aged battery compared to that of the fresh battery and that the size of the deposits increases with decreasing SOH. This means increased growth of the CEI with increased cycling at low temperature. The results of the EDS tests are shown in Fig. S5, Table S1, and Table S2. Table S1 shows that the relative atomic percent of C decreases after cycling. However, the percents of O, F, P, S, Ni, Co, and Mn increase. This means that deposits are formed on the surface of the anode. The occurrence of Ni, Co, and Mn on the surface of the anode indicates transition metal dissolution, which increased with decreasing SOH during low-temperature cycling. This means that lithium plating accelerates transition metal dissolution.

Fig. S5(l–p) show that more deposits cover the surface of the battery anode with decreasing SOH. Also, the element distribution of the bumps

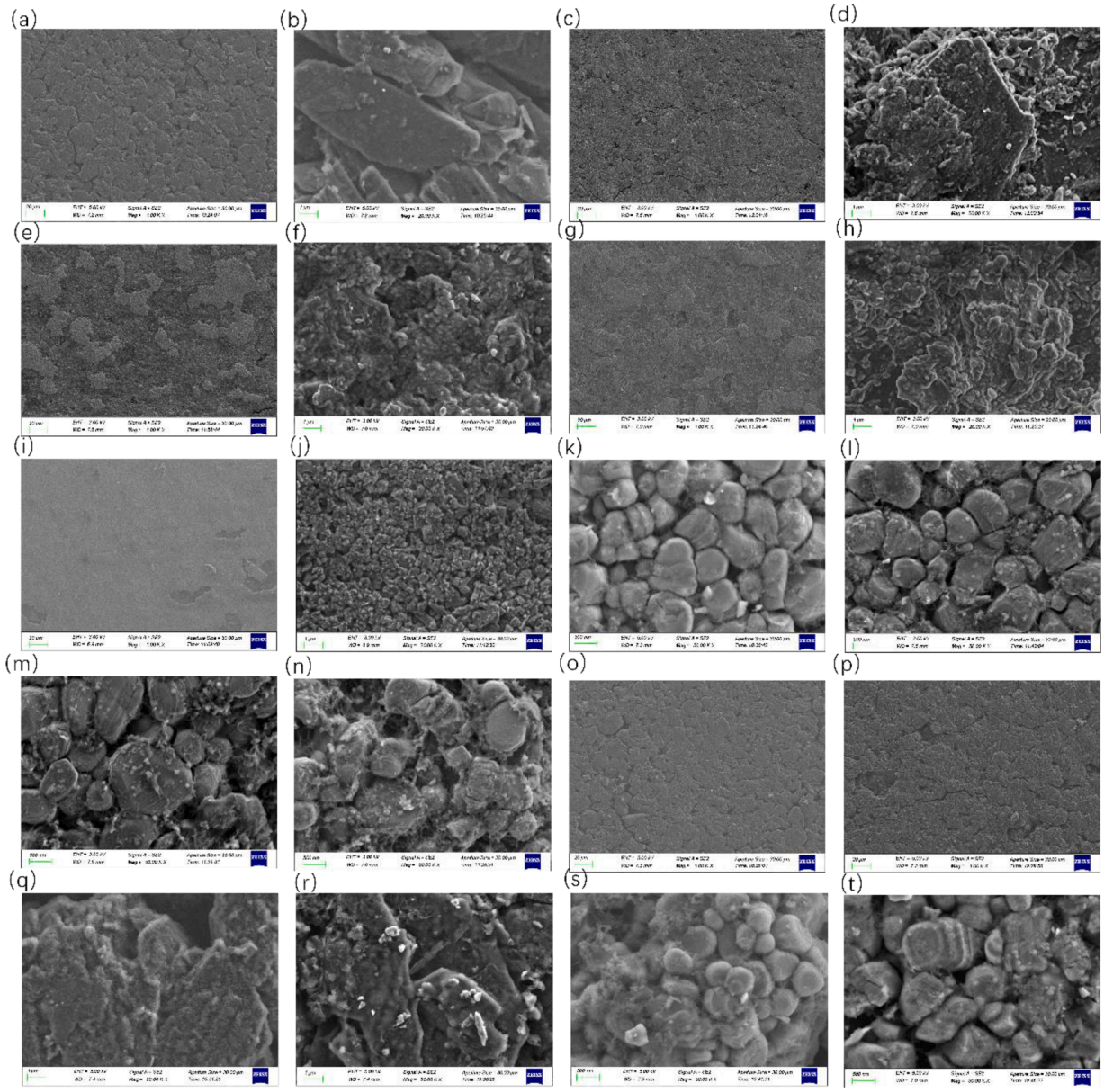


Fig. 2. SEM images of battery aging. (a),(b) Anode of fresh battery (1,20 KX); (c-f) Anode of battery with 98.7%,90% SOH aging at −10 °C; (1,20KX); (g-j) Anode of battery with 70% SOH aging at −10 °C (part 1–1 KX,20 KX, part 2–1 KX, 20 KX); (k)Cathode of fresh battery (50 KX); (l-n) Cathode of battery with 98.7% SOH,90% SOH, 70% SOH aging at −10 °C (50 KX); (o) Anode of battery with 70% SOH aging at 50 °C; (1 KX); (p-r) anode of battery with 70% SOH aging at 50 °C after −10 °C (part 1–1 KX,20 KX, part 2–20 KX); (s) cathode of battery with 70% SOH aging at 50 °C; (50 KX); (t) cathode of battery with 70% SOH aging at 50 °C after −10 °C (50 KX).

is the same as that of anode of the battery with 70% SOH (part 2). Electrolyte decomposition increases with increased cycling at low temperature. Table S2 shows less change in the elements on the surface of the cathode. This means that fewer side reactions occurred at the surface of cathode except for transition metal dissolution.

To analyze the elemental constituents of the deposits on the surface of the anode, XPS analysis was conducted, and the results are shown in Fig. 3. It is shown in Fig. 3(a) that four peaks are detected in the C 1 s spectra of the anodes of batteries with different SOH values. The dominant peak at 284.5 eV is attributed to graphite. The main peaks at approximately 286.8, 288.3, and 289.8 eV are characterized by C–C, C=O, and CO_3 bonding, respectively [37–41]. It can be seen from Fig. 3

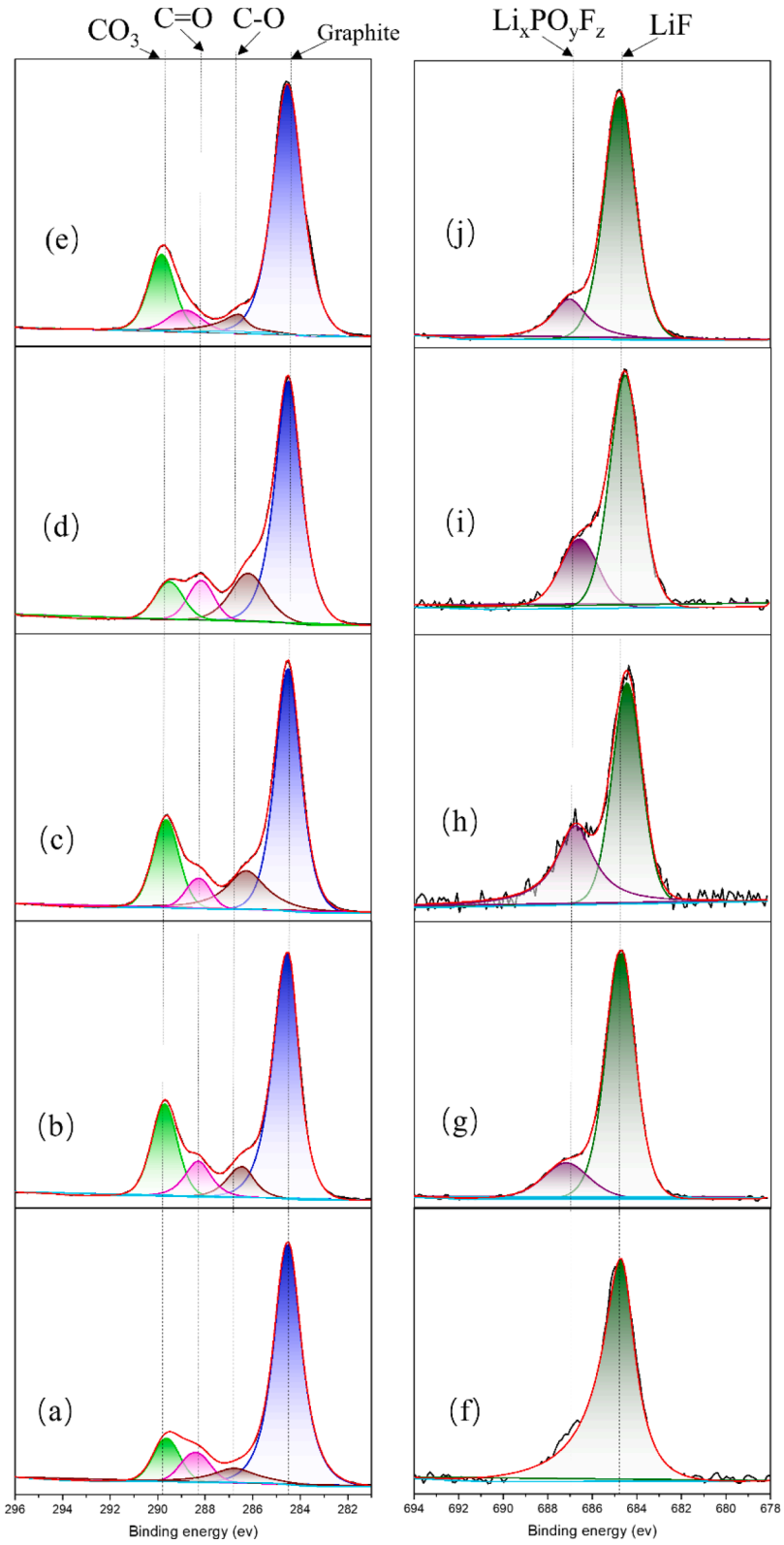


Fig. 3. XPS tests results of batteries. (a) C 1s: Anode of fresh battery; (b) C 1s: Anode of battery with 98.7% SOH aging at −10 °C; (c) C 1s: Anode of battery with 90% SOH aging at −10 °C (part 1); (d) C 1s: Anode of battery with 70% SOH aging at −10 °C (part 1); (e) C 1s: anode of battery with 70% SOH aging at 50 °C after −10 °C (part 1); (f) F 1s: Anode of fresh battery; (g) F 1s: Anode of battery with 98.7% SOH aging at −10 °C; (h) F 1s: Anode of battery with 90% SOH aging at −10 °C (part 1); (i) F 1s: Anode of battery with 70% SOH aging at −10 °C (part 1); (j) F 1s: anode of battery with 70% SOH aging at 50 °C after −10 °C (part 1).

(a–d) that the C–O and C=O peaks intensity increases with decreasing SOH. However, the CO_3 peak intensity initially increases with increasing SOH and then decreases. This means that inorganic (Li_2CO_3) deposition is the main constituent of the initial deposits on the surface of anode. As lithium plating progresses with extended CV time, electrolyte decomposition begins and generates organic deposits (C–O and C=O). These organic deposits then become the main constituents on the anode surface.

There are two peaks in the F 1 s spectra, shown in Fig. 3(g–i). The peaks at approximately 684.8 and 687.6 eV are attributed to Li–F and $Li_xPO_yF_z$, respectively [39,40,42]. It can be seen from Fig. 3(f) that only the peak at 684.8 eV occurs. $Li_xPO_yF_z$ initially increases with increased cycling and then decreases. The XPS results for part 2 of the anodes are depicted in Fig. S6. The constituents change of the deposit on the surface of anode at part 2 are in accordance with that at part 1. However, it can be seen from Fig. S6(a–c) and Fig. 3(b–c) that the organic constituents of the deposits in part 2 of the anode are much greater than those of part 1. The peak for CO_3 even disappears from the surface of anode at part 2 for the battery with 70% SOH. This means that organic compositions are the primary products at the second stage for battery aging during low-temperature cycling. It is also concluded that lithium plating is inhomogeneous and that the kinks and edges of the "jelly roll" are prone to lithium plating.

To study the change in crystal structure of the electrodes of the aged batteries, XRD analysis was conducted. The results for the cathode and anode are shown in Fig. 4(a) and (b), respectively. The results shown in Fig. 4(a) indicate that there was less change in the crystal structure of the cathode of the battery aged at low temperature compared to fresh battery. However, Fig. 4(b) shows a new peak in the XRD curve of the anode of the battery with 70% SOH aging at low temperature. The new peak represents LiC_{20}. This means that lithium plating formed continuously after cycling at low temperature.

Based on the experimental results, it is concluded that battery aging is divided into two stages and that SOH decreases linearly with increased cycling in both stages during low-temperature cycling. At the first stage, "dead lithium" is the primary aging mechanism, and the constituent of the deposition at the surface of anode is mainly inorganic. At the second stage, SEI growth is the primary aging mechanism, and extended CV mode causes serious side reactions. The constituents of the deposits at the surface of anode are mainly organic in the second stage. However, "dead lithium" causes more battery aging than does SEI growth. The plated lithium on the anode surface is inhomogeneous. The kinks and edges of the "jelly roll" are prone to lithium plating, and more lithium plating accelerates electrolyte decomposition. The plated lithium and products are present as white bumps on the surface of the anode. Except for transition metal dissolution, there is less cathode aging under low-temperature cycling.

3.2. Battery aging at high temperature after low-temperature cycling

After the SOH for battery aging at −10 °C decreased to 90%, the aged lithium-ion batteries were cycled at 50 °C to study battery aging behavior and mechanisms at high temperature after low-temperature cycling, and the results are shown in Fig. 5. It can be seen from Fig. 5 (a) that the battery aging at high temperature is accelerated after low-temperature cycling because when the SOH of the battery decreases from 90% to 70% under constant high-temperature cycling, the cycle number is more than two hundred (shown in Fig. S7). The battery aging acceleration factor, *AF*, is obtained from Eq. (2).

$$AF = \frac{CN_{\mathrm{h}}}{CN_{lh}} \tag{2}$$

where CN_h is the cycle number when the SOH for the battery decreases from SOH_1 to SOH_2 at a constant high temperature, and $CN_{l\text{-}h}$ is the cycle number when the SOH for the battery decreases from SOH_1 to SOH_2 at high temperature after low-temperature aging. The calculated results show that *AF* is approximately 2.1.

Fig. 5(b–d) show that the change of IC curves with increasing cycle number for lithium-ion batteries aged at high temperature after low-temperature cycling is similar to that of the battery aged at constant low temperature. Peak (1) decreases significantly after 90% SOH, which means more LAM at the anode. Also, the rate of decrease of peak (1) for the battery aged at constant high temperature, shown in Fig. 5(d), is much slower than that of the batteries shown in Fig. 5(b) and (c). Peak (1) in Fig. 5(d) does not shift as quickly as peak (1) in Fig. 5(b–c). This means that LLI shown in Fig. 5(d) is less than that shown in Fig. 5(b–c) [16]. Besides, peaks (1) and (2) in Fig. 5(b–c) shift to higher voltages compared to peaks (1) and (2) in Fig. 5(d). However, peak (2) in Fig. 5 (b) decreases less than peak (2) in Fig. 5(c–d). Peak (2) represents cathode aging. Because the cathode aging shown in Fig. 5(b) is less than that shown in Fig. 5(c–d) for batteries with the same SOH, this means that anode aging is more serious for the battery shown in Fig. 5(b). The EIS plots are shown in Fig. 5(e). It can be seen that the two semicircles are significantly larger after cycling at high temperature compared with the EIS plots shown in Fig. 1(e), which means that serious side reactions occurred and many deposits covered the surface of the battery anode. This was caused by the deposition of more plated lithium in the battery cycled at constant low temperature.

To obtain more information about the aging mechanisms for lithium-

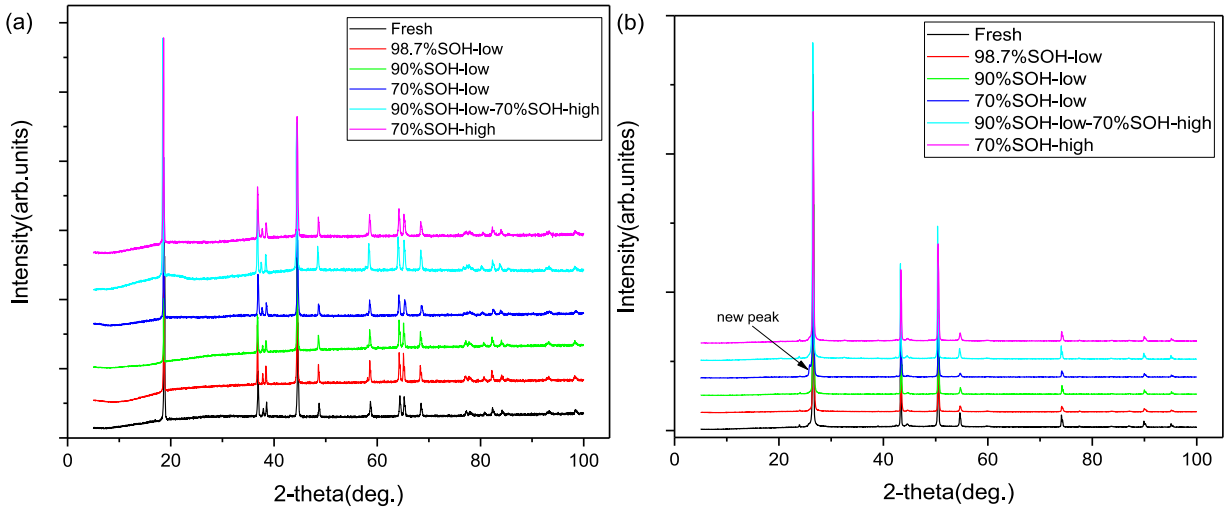


Fig. 4. The results of XRD test. (a) cathode; (b) anode.

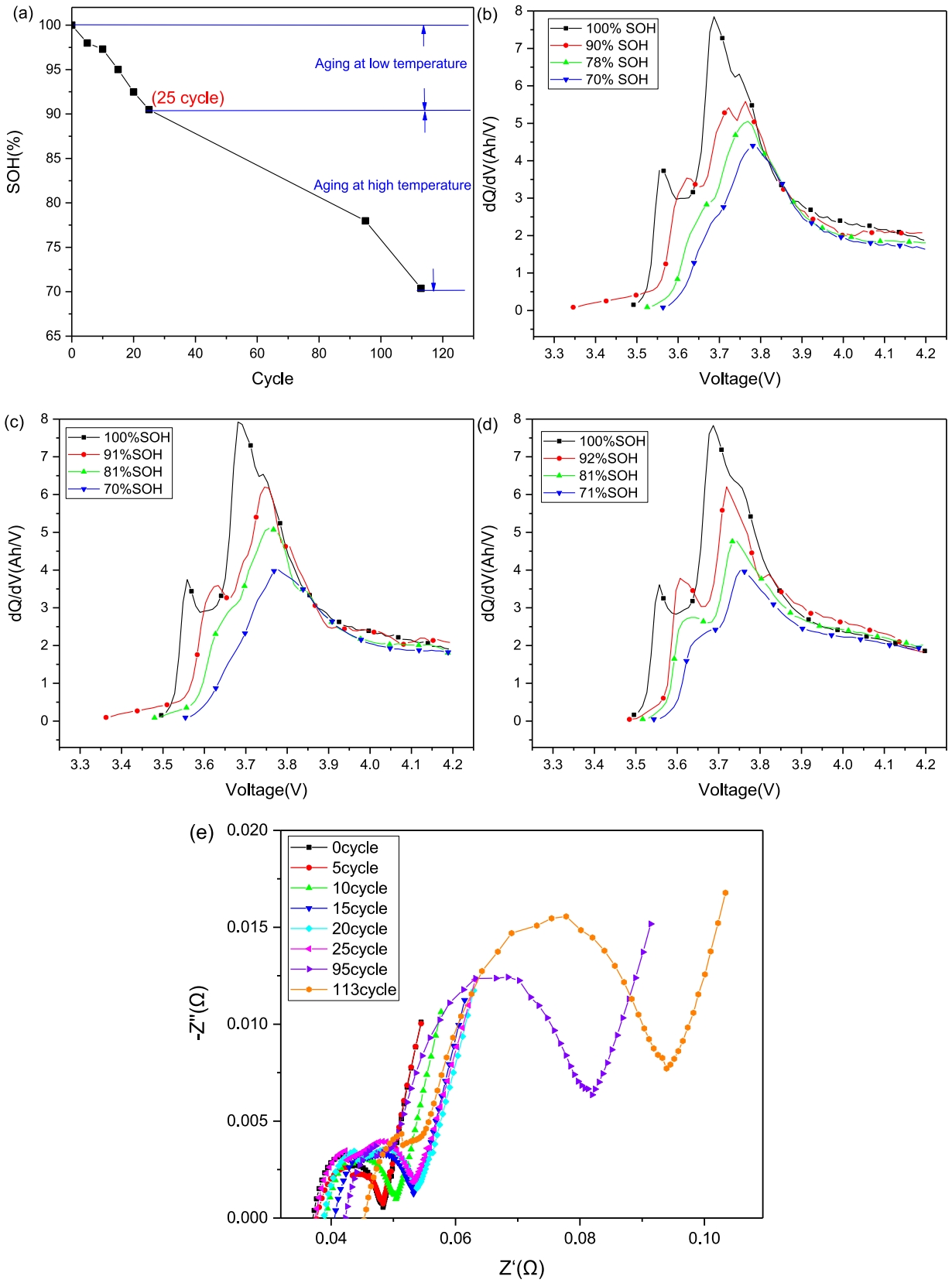


Fig. 5. Battery aging at high temperature after low temperature. (a)SOH changes with cycling; (b) IC curves from capacity test of battery aged at 50 °C after −10 °C; (c) IC curves from capacity test of battery aged at −10 °C; (d) IC curves from capacity test of battery aged at 50 °C; (e) EIS curves of battery aged at 50 °C after −10 °C.

ion batteries, SEM, EDS, XPS, and XRD tests were conducted to study the changes in surface topography, elements, constituents, and crystal structure, respectively, for the aged battery. The optical image shown in Fig. S2 for the battery with 70% SOH aging at 50 °C after being aged at −10 °C is similar to that of the battery with 70% SOH aged at a constant −10 °C except that lithium plating is not found for the high-temperature-aged battery. The results of the SEM analysis are shown in Fig. 2(o–t). Fig. 2(o) and (q) show that the anode of the battery with 70% SOH at a constant high temperature retains its sheet structure, and some deposits cover the surface of the anode. However, more deposits cover the surface of the anode for the battery with 70% SOH aging at 50 °C after −10 °C aging, as shown in Fig. 2(p) and (r). This means that serious side reactions occurred. Compared to Fig. 2(c) and (d), no bumps are visible in Fig. 2(p). Also, Fig. 2(r) shows that fewer products cover the surface of the anode compared to the anodes shown in Fig. 2(f) and (h). Although deposits cover the surface of the anode shown in Fig. 2(r), the sheet structure can still be seen. There are few needle-like products on the surface of anode shown in Fig. 2(e), which means that no new lithium plating occurred during the high-temperature cycle. It can be seen from Fig. S8 that there is no new peak on the IC curves derived from discharging at high temperature. Lithium plating therefore accelerates electrolyte decomposition during high-temperature cycling. Deposits cover the surface of the cathode shown in Fig. 2(s). However, there was less change to the cathode shown in Fig. 2(t) compared to that of the fresh battery shown in Fig. 2(k). This indicates more significant formation of a CEI for the cathode of the battery aged at a constant high temperature. Anode aging is more serious than cathode aging for batteries aged at high temperature after low-temperature aging. The occurrence of lithium plating changes the aging mechanisms of batteries aged at a constant high temperature.

The EDS results are shown in Fig. S4(i), (j), and (q), Table S1, and Table S2. Fig. S4(q) shows that the element distribution on the surface of the anode is uniform. Table S1 shows that the relative atomic percent of C increases. However, those of elements O, F, P, S, Ni, and Mn increase. This means that side reactions occurred on the surface of the anode. High-temperature cycling did not accelerate transition metal dissolution when comparing the results for the battery with 90% SOH at low temperature with those of the battery with 70% SOH aging at high temperature after low-temperature aging. Table S2 shows less change in the relative atomic percents of Ni, Co, and Mn for the battery with 70% SOH aging at high temperature after low-temperature aging. However, the relative atomic percent of these elements for the battery with 70% SOH aging at high temperature decreased. These results are consistent with those of the SEM analysis.

The XPS results for battery aging at high temperature after low-temperature aging are shown in Fig. 3(e) and (j) and Fig. S6(d) and (h). It is shown that there is a greater amount of inorganic constituent ($LiCO_3$) for parts 1 and 2 of the battery aged at high temperature after low-temperature aging than of organic constituents (C–O and C=O).

Compared to the XPS spectra shown in Fig. 3(e) and Fig. S6(d), it was found that the relative content of the organic constituent in part 2 was greater than that in part 1. Plated lithium was more prevalent in the kinks and edges of the "jelly roll." This explains why the organic constituent in part 2 was greater than that than part 1. Fig. 3(j) and (i) and Fig. S6(g) and (h) show that the relative content of LiF for the anode surface of the battery with 70% SOH aging at high temperature after low-temperature aging is greater than that of the battery with 70% SOH aging at low temperature. The XRD analysis results shown in Fig. 4(a) and (b) show that there was less change to the crystal structures of the cathode and anode for the battery aged at high temperature after low-temperature aging.

Based on the above results, it is concluded that plated lithium accelerates battery aging under high-temperature cycling and that the occurrence of plated lithium changes the battery aging mechanisms under constant high-temperature cycling. LAM at the anode increased for the battery cycled at high temperature after low-temperature aging compared to that of the battery cycled at constant temperature. More side reactions occurred on the surface of the anode for the former. It is also noted that plated lithium disappeared after cycling at high temperature, indicating its consumption. It is thought that some of it was intercalated into the anode as reversible lithium and that the rest reacted with the electrolyte because there were fewer deposits covering the surface of the anode of the battery cycled at high temperature after low-temperature aging than that of the battery cycled at constant low temperature. Plated lithium accelerates electrolyte decomposition and generates organic products. No new lithium plating at high temperature is reported in this paper. Inorganic products became the main components of the deposits on the surfaces of the anodes, and SEI growth is the primary aging mechanism. Transition metal dissolution at high temperature cycle did not accelerate.

3.3. *Influence of factors on battery aging at high temperature after low-temperature aging*

The plated lithium content increased as the SOH decreases at constant low-temperature cycling. In these experiments, lithium-ion batteries were cycled to a predetermined SOH at low temperature. They were then cycled at high temperature to study the influence of plated lithium content on battery aging at high temperature after low-temperature aging.

3.3.1. *Plated lithium content*

The results for battery aging are shown in Fig. 6. It can be seen from Fig. 6(a) that the battery aging rate increased with decreasing SOH. The acceleration factors for batteries with 97, 90. and 82% SOH were approximately 1.5, 2.1, and 4.7, respectively. This means that plated lithium accelerated battery aging with high-temperature cycling.

The IC curves for batteries with 70% SOH are shown in Fig. 6(b–c). The change in the IC curves shown in Fig. 6(b) is similar to that shown in Fig. 5(d). There is less change to peaks (1) and (2) in Fig. 6(b) compared to those in Fig. 5(d). The decreased battery cycle life shown in Fig. 6(b) is caused by the fast side reaction rate. The IC curves shown in Fig. 6(c) and Fig. 5(b) are similar to that shown in Fig. 5(c). The difference is that the shift in peak (1) in Fig. 6(c) and 5(b) is slower than that in Fig. 5(c). Lithium plating is continuous under low-temperature cycling. However, it is consumed under high-temperature cycling. The continuous lithium plating and increased SEI film growth caused more LLI, as shown in Fig. 5(c). Peak (2) in Fig. 6(c) shows less change compared to that in Fig. 5(c) at the same SOH. Aging for the battery shown in Fig. 6(c) is more similar to that of the battery aged at constant low temperature. It can be seen from Fig. 6(a) that the aging rate for batteries with SOH values higher than 82% at low temperature decreases after cycling at high temperature. However, there was less of a change in aging rate for the battery with 82% SOH at low temperature after cycling at high temperature. This supports the conclusion that the increase in plated lithium caused more side reactions and accelerated battery aging. The influence of plated lithium content on the aging mechanisms for batteries under high-temperature cycling changed with increasing content. At first, the content of deposited lithium was restricted and depleted quickly under high-temperature cycling. Although this accelerated battery aging, the aging mechanisms are still similar to those for battery aging under extended high-temperature cycling. As the amount of deposited lithium increased, more serious side reactions occurred. Then, the aging mechanisms became similar to those for battery aging at low temperature. As the amount of deposited lithium content increased further, very serious side reactions occurred. The aging rate of a battery after high-temperature cycling thus changes less than that of a battery after low-temperature cycling.

The EIS results are shown in Fig. S9(a) and (b). It can be seen that the two semicircles in the mid-frequency regions shown in Fig. 5(e) expands more than those shown in Fig. S9(a) and (b). The increased plated lithium content accelerated side reactions, which increased SEI and

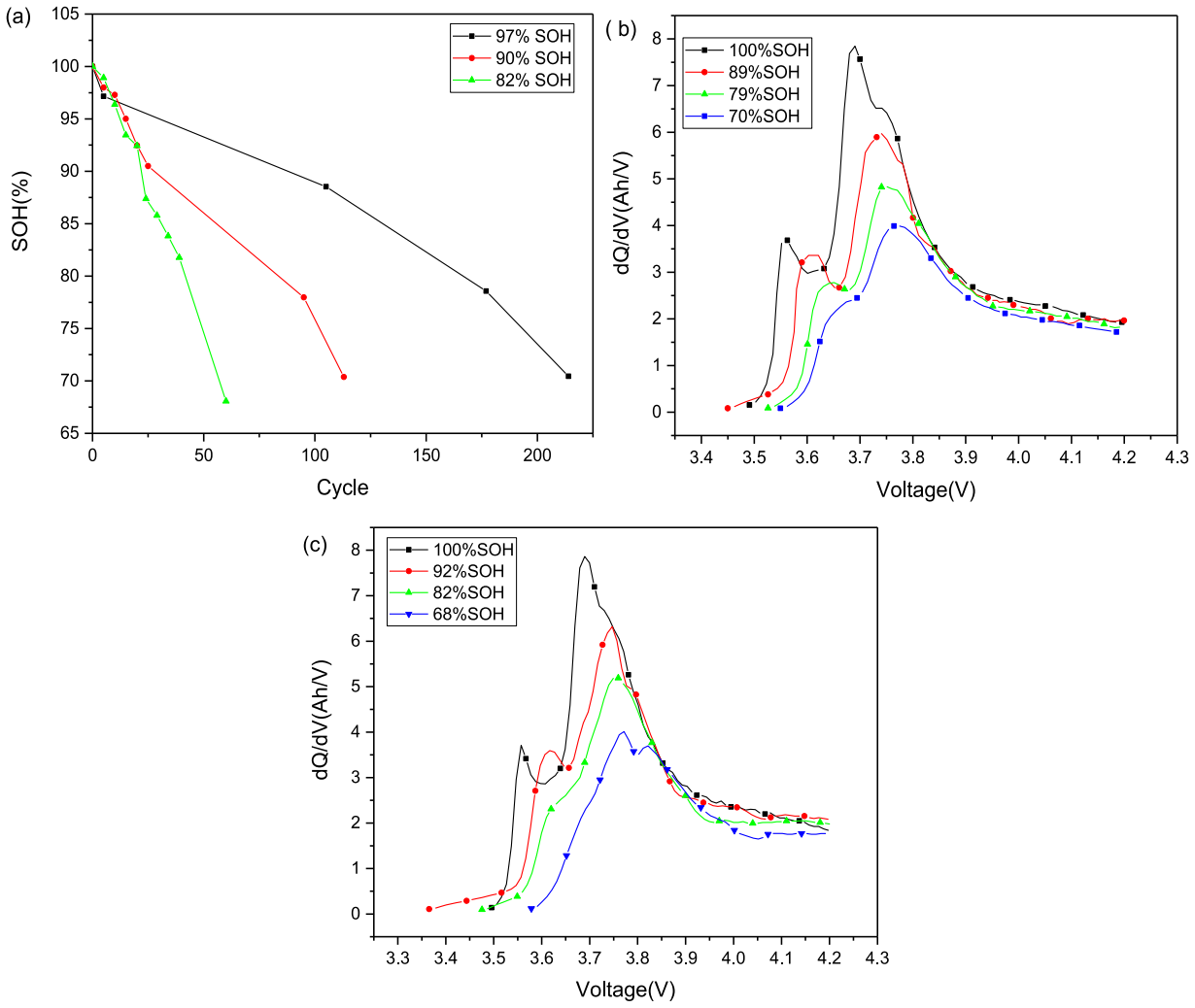


Fig. 6. Battery with different SOH aging at 50 °C after −10 °C. (a)SOH changes with cycling; (b) IC curves from capacity test of battery with 97% SOH; (c) IC curves from capacity test of battery with 82% SOH.

charge-transfer resistance. However, in this case, plated lithium was not depleted as its content increased. The occurrence of plated lithium caused the resistance increase rate decrease. It can be also seen from Fig. 1(e) that the two semicircles expanded least compared to the other batteries. Because peak (1) represents the final lithium interaction with the anode, the greater decrease for peak (1) in Fig. 6(c) means that the plated lithium content that reacted with the electrolyte was greater than that intercalated into the anode.

3.3.2. *Charging rate*

To study the influence of charging rate at high temperature on battery aging at high temperature after low-temperature cycling, batteries were cycled to 90% SOH at low temperature. They were then cycled at high temperature with 0.5, 1, and 1.5C CC–CV charging and 1C CC discharging. The results of battery aging with different charging rates at high temperature are shown in Fig. 7. It can be seen from Fig. 7(a) that lithium-ion battery aging accelerated at high temperature after low-temperature aging because batteries can be cycled more than one hundred times when they are cycled from 90% to 70% SOH. The acceleration factors for batteries charging with 0.5, 1, and 1.5C were 4.6, 2.1, and 5.9, respectively. It is interesting that the acceleration factors initially decreased with increased charging rate and then increased.

The IC curves for the battery with 70% SOH are shown in Fig. 7(b) and (c). The change in the IC curve shown in Fig. 7(b) is similar to that shown in Fig. 5(c). The difference is that peak (1) shifts faster and to a higher voltage in Fig. 7(b) than in Fig. 5(c) in the range from 70 to 80% SOH. It can be also seen that peak (2) decreases less and shifts to higher voltage. These results indicate that LLI for the battery shown in Fig. 7(b) is greater than that for the battery shown in Fig. 5(c). Thus, serious anode aging caused fast battery aging. Although low charging means slow side reactions, long cycling caused the accumulation of additional products for the reactions between plated lithium and the electrolyte. The result of the EIS test for the battery charged with 0.5C is shown in Fig. S9(c). It can be seen that the semicircles in the mid-frequency regions of the EIS plots expand the most compared to other batteries. This also indicates additional deposits on the surface of the anode. Thus, serious anode aging caused battery aging. Plated lithium was depleted owing to the greater number of cycles.

The IC curve change shown in Fig. 7(c) is similar to that shown in Fig. 5(d). The difference is that peaks (1) and (2) shift faster and to higher voltages than the peaks shown in Fig. 5(d). However, peaks (1) and (2) decrease slower than those shown in Fig. 5(d). This means that LLI of the battery shown in Fig. 7(c) is more serious. Because peak (1) reflects the final lithium interaction with the anode, it is expected that plated lithium intercalated into the anode. Thus, peak (1) decreases less than it does in Fig. 5(d). The results of the EIS test for battery charging with 1.5C is shown in Fig. S9(d). The two semicircles expand less when the SOH of the battery decreases to 70% SOH. Although high charging rates accelerated SEI growth, plated lithium was not depleted owing to the limited number of cycles. The EIS change shown in Fig. S9(d) is

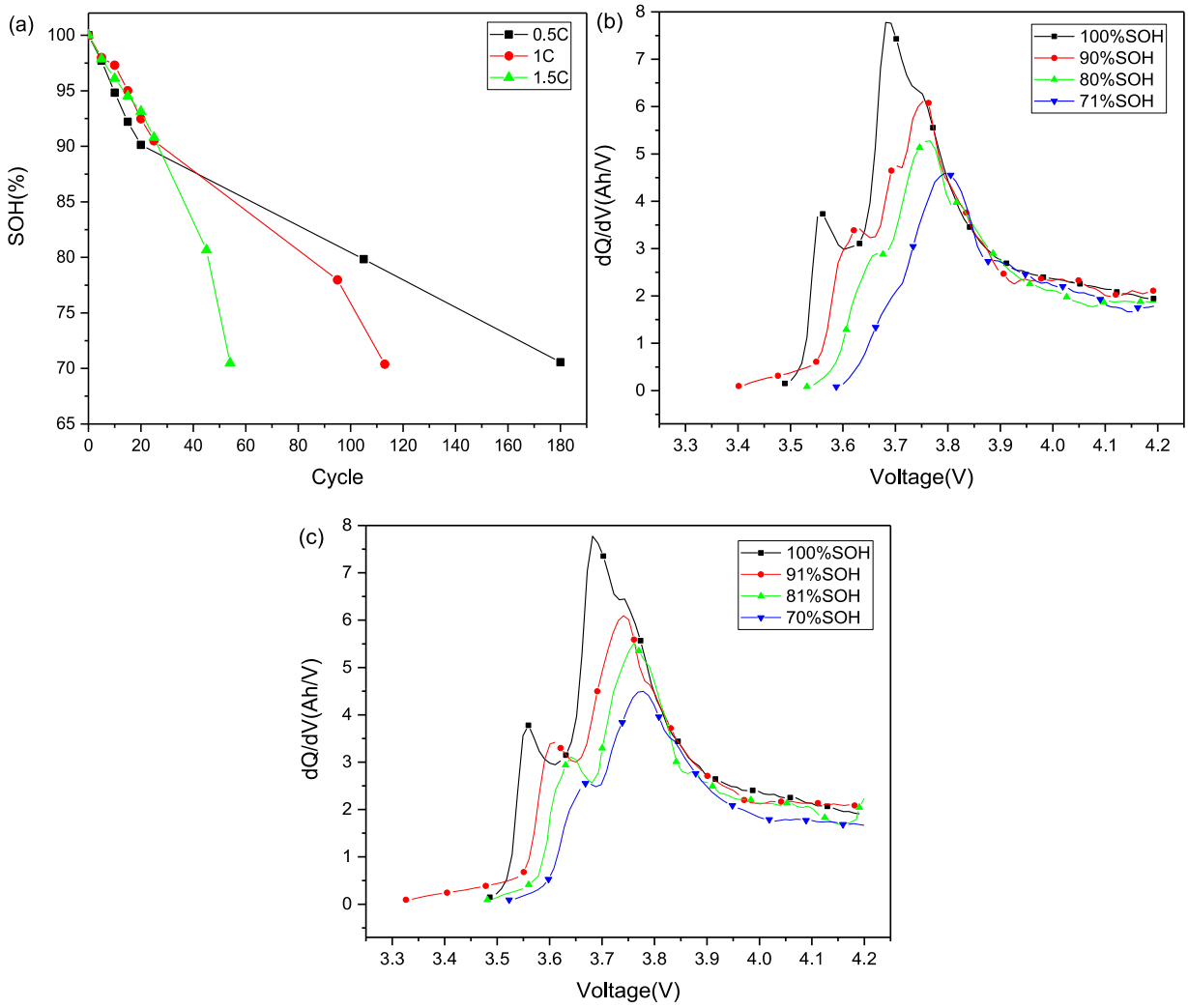


Fig. 7. Batteries aging with different charging rate at high temperature after low temperature. (a)SOH changes with cycling; (b) IC curves from capacity test of battery charging with 0.5 C cycle; (c) IC curves from capacity test of battery charging with 1.5 C cycle.

similar to that shown in Fig. 1(e). The optical image and XPS results for the battery cycled at high temperature with 1.5C after low-temperature cycling are shown in Fig. S2 and Fig. S6(i–l), respectively. It can be seen from Fig. S2 that the battery charged at 1.5C is similar to the battery with 70% SOH at constant low temperature. Plated lithium can be seen on the anode. Fig. S6(i) shows a greater amount of organic products at part 1 for the battery charged with 1.5C than inorganic products. Fig. S6 (k) shows that the peak for CO_3 disappears at part 2 on the surface of the anode for the battery charged with 1.5C. This is similar to the results for the battery with 70% SOH at constant low temperature. This also means that plated lithium was not depleted for the battery cycled at high temperature with 1.5C after low-temperature cycling. Peak (1) in Fig. 5 (c) shifts faster and more to a higher voltage than that in Fig. 7(c). Peak (1) in Fig. 5(c) decreases faster than that in Fig. 7(c). This means that LLI and anode aging shown in Fig. 5(c) are more serious than that shown in Fig. 7(c). However, the two semicircles in Fig. S9(d) expand more than those in Fig. 1(e). It is expected that “dead lithium” and SEI growth are the main aging mechanisms on the anode for the battery shown in Fig. 5 (c). SEI growth is thus more serious for the batteries shown in Fig. 7(c) and Fig. S9(d). The slower decrease of peak (1) shown in Fig. 7(c) is caused by plated lithium intercalating into the anode. The IC curves for battery aging at constant high temperature with 1.5C are shown in Fig. S10(a). Compared with the results shown in Fig. 5(d), peak (1) shifts faster and to a higher voltage, and peak (1) decreases faster. However, the change for peak (2) shown in Fig. S10(a) is similar to that shown in Fig. 5(d). This means that fast charging mainly accelerates SEI growth. The greater expansion of the two semicircles for the battery charged with 1.5C at a constant high temperature shown in Fig. S10(b) also supports this conclusion.

3.3.3. *Discharging rate*

To study the influence of discharging rate at high temperature on battery aging at high temperature after low-temperature cycling, batteries were cycled to 90% SOH at low temperature. They were then cycled at high temperature with 1C CC–CV charging and 0.5, 1, and 1.5C discharging. The results of battery aging behavior with different charging rates at high temperature are shown in Fig. 8(a). Compared to the results shown in Fig. 7(a), the aging rate for the battery charged with 1.5C is faster than that of the battery discharged with 1.5C. However, the aging rate for the battery charged with 0.5C is slower than that of battery discharged with 0.5C. It is shown in Fig. 8(a) that the aging rate with high-temperature cycling initially decreases with increase discharging rate and then increases.

The IC curves for the batteries with approximately 70% SOH are shown in Fig. 8(b) and (c). It can be seen from the figures that the change in peak (1) is similar to that shown in Fig. 5(d). The difference is that peaks (1) and (2) in Fig. 8(b) and (c) move faster and to higher voltages. This means that LLI for the battery shown in Fig. 8(b) and (c) is greater than that for the battery shown in Fig. 5(d). Electrolyte decomposition is thus more serious for the battery shown in Fig. 8(b) and (c). The EIS plots

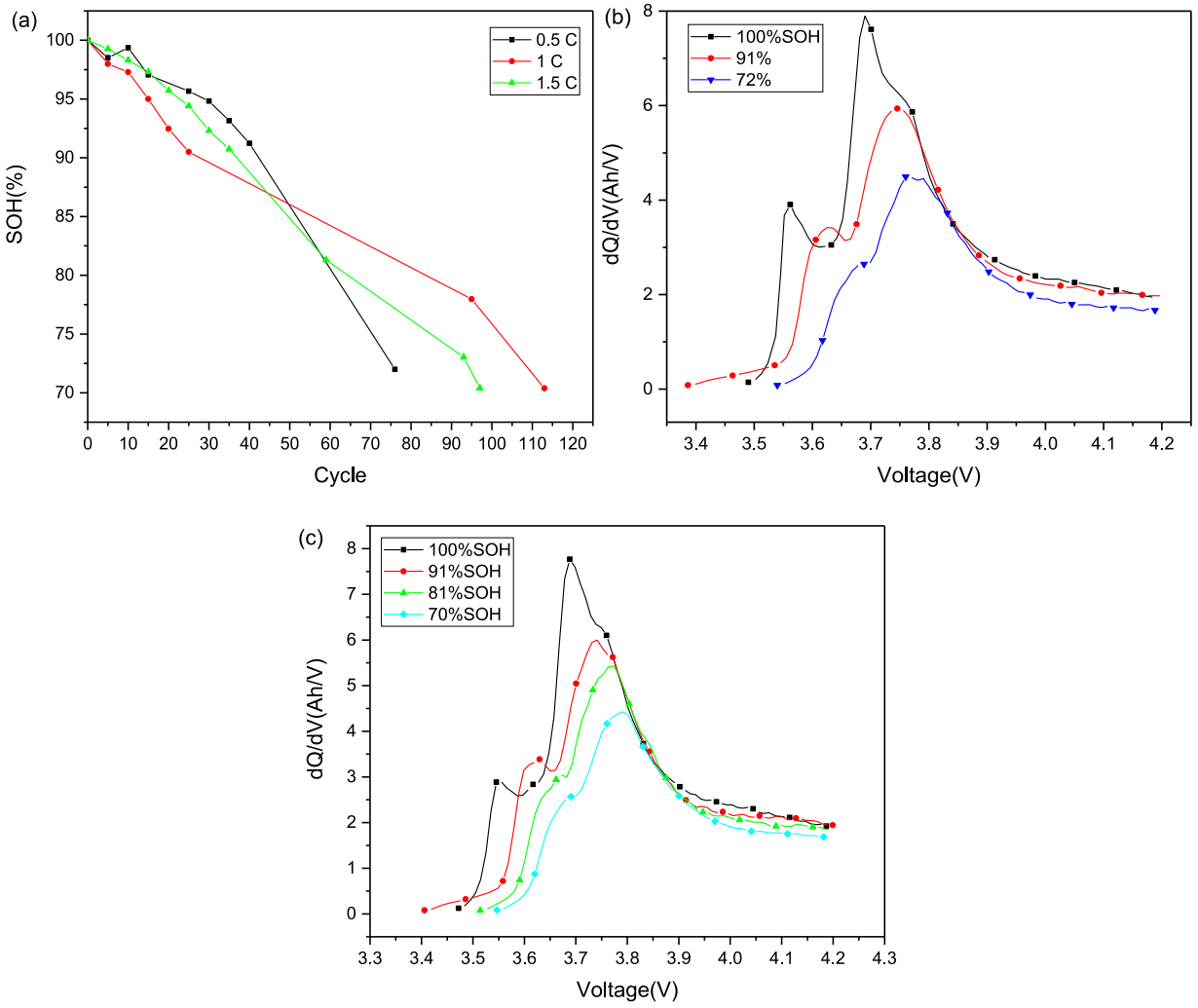


Fig. 8. Batteries aging with different discharging rate at high temperature after low temperature. (a)SOH changes with cycling; (b) IC curves from capacity test of battery discharging with 0.5 C cycle; (c) IC curves from capacity test of battery discharging with 1.5 C cycle.

shown in Fig. S7(e) and (f) indicate greater expansion of the two semicircles, which indicates more serious side reactions and SEI growth. However, peaks (1) and (2) decrease slower for the batteries shown in Fig. 8(b) and (c) because the plated lithium is intercalated into the anode. Compared to the results shown in Fig. 5(b), peak (1) in Fig. 8(b) and (c) decreases slower, which is caused by the plated lithium not being depleted owing to the limited number of cycles.

When the discharging rate was 0.5C, the discharging time increased. The organic products generated during charging decreased and the number of inorganic products increased at higher temperature, causing more side reactions and accelerating battery aging. When the discharging rate was 1.5C, the thermal effects of the discharging rate increased compared to that at 1C [27], increasing the number of inorganic products. This also accelerated battery aging compared to the discharging rate of 1C. A high discharging rate also caused more discharging stress in the materials. High temperature thus had more of an effect on battery aging at high temperature after low-temperature cycling than discharging rate.

Based on the experimental results reported in this section, the primary aging mechanisms for battery aging at high temperature after low-temperature cycling are electrolyte decomposition and SEI growth. Plated lithium is consumed during high-temperature cycling, with some of it intercalating into the anode and the rest reacting with the electrolyte to generate SEI. Plated lithium content and charging–discharging rate at high temperature influence plated lithium consumption. The effect of plated lithium content at high temperature after low-temperature cycling is divided into three stages. In the first stage, plated lithium accelerates aging and electrolyte decomposition, and it is depleted over time. The aging mechanisms are similar to those of batteries cycled at constant high temperature. In the second stage, the higher plated lithium content causes more serious side reactions, but it is still depleted owing to the relatively long cycle. Peak (1) in the IC curves decreases and shifts faster to a higher voltage compared to peak (1) for the battery cycled at high temperature. In this stage. anode aging is more serious. In the third stage, the presence of plated lithium causes much more serious side reactions and it is not depleted. Peak (1) decreases less as plated lithium is intercalated into the anode. However, peak (1) shifts faster and to higher voltages, causing increased LLI. Although a low charging rate means slower side reactions, the accumulated products resulting from side reactions during extended cycling cause more serious anode aging. Plated lithium is depleted, and a high charging rate accelerates electrolyte decomposition and SEI growth. However, plated lithium is not depleted owing to limited number of cycles. A low discharging rate means more discharging time, and a high temperature causes more side reactions. The thermal effects of discharging rate increase for high discharging rates; it also causes more side reactions. However, plated lithium is not depleted. Peak (1) decreases slower for them owing to the intercalation of plated lithium into anode, although LLI increases more. The impact of these factors on battery aging at high temperature after low-temperature cycling in order of importance is

charge current > plated lithium content > high temperature > discharge current.

4. Conclusions

In summary, the aging behavior and mechanisms for batteries cycled at high temperature after low-temperature cycling and the influence of three factors on the aging behavior were investigated. First, the plated lithium content increased with cycling at low temperature. Battery aging at a constant low temperature was divided into two stages, and "dead lithium" was the primary aging mechanisms at the first stage. SEI growth was the primary aging mechanisms at the second stage, whereas "dead lithium" caused more capacity fading. The results for batteries cycled at high temperature after low-temperature cycling indicated that plated lithium was consumed during high-temperature cycling. Some of the plated lithium was intercalated into the anode, and the rest reacted with the electrolyte to generate SEI. The occurrence of plated lithium at high temperature after low-temperature cycling accelerated battery aging compared to a battery aged at a constant high temperature. The products of electrolyte decomposition on the surface of the anode were mainly organic at low temperature when the SOH of battery decreased to 70%. However, the products for electrolyte decomposition were mainly inorganic at high temperature after low-temperature cycling. Electrolyte decomposition and SEI growth were the primary aging mechanisms for battery aging at high temperature after low-temperature cycling. Increased plated lithium content caused more serious side reactions at high temperature after low-temperature cycling, which affected battery aging. However, the plated lithium was not depleted as it increased to the threshold. High charge rates accelerated side reactions, but again, plated lithium was not depleted owing to the limited number of cycles. Low discharging rates mean more discharging time at high temperature, and thus the thermal effects of discharging rate increased for high discharging rates. High temperatures caused more side reactions. However, plated lithium was not depleted owing to the limited number of cycles. The impact of these factors on battery aging at high temperature after low-temperature cycling in order of importance is charge current > plated lithium content > high temperature > discharge current.

However, this research also has some shortcomings. In the future work, we will conduct experiments to study the influence of CV time on the lithium plating to search for better charge method at low temperature. Besides, the battery cycled at low temperature will cycled under slight overcharging, because aged lithium-ion battery is prone to slight overcharging. The safety and cycle life issues will be discussed then.

CRediT authorship contribution statement

Jialong Liu: Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing. **Yun Zhang:** Methodology, Investigation. **Jinlong Bai:** Methodology, Investigation. **Longfei Zhou:** Investigation. **Zhirong Wang:** Supervision, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.electacta.2023.142362.

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